Hyeongmeen Baik

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EDUCATION

University of Wisconsin - Madison, Madison, Wisconsin — Aug 2023 — Expected 2028
Ph.D., Electrical & Computer Engineering
Wisconsin Electric Machines and Power Electronics Consortium (WEMPEC)
Advisor: *Prof. Jinia Roy*

Yonsei University, Seoul, Korea — Mar 2017 — Feb 2023
B.S., Electrical & Electronic Engineering
Thesis: Short-term Photovoltaic Power Forecasting Model, based on Artificial Neural Network and Meteorological Factors
Advisor: *Prof. Jung-wook Park*

RESEARCH EXPERIENCE

WEMPEC, University of Wisconsin - Madison — Madison, WI
Research Assistant (Advisor: *Prof. Jinia Roy*) — Aug 2023 - Ongoing

- **Active capacitor converters:** Analyzed power pulsation buffer (PPB) and active capacitor control for high peak-to-average pulsed loads, and presented a 90-min peer-reviewed tutorial on active capacitor circuits and control at IEEE ITEC+EATS 2026.
- **Renewable:** Built a GaN three-phase interleaved buck converter as a DC-biased AC excitation source for in-situ PEM electrolyzer impedance measurement, decoupling voltage regulation from current sharing, and measured impedance matched a laboratory potentiostat within 5%. Designed a multi-port extended-duty-ratio boost converter for hybrid power sources.
- **Pulsed power:** Replaced hysteresis control of a 768 V / 982 A IGBT flashlamp driver with fixed 20 kHz dual-mode loop shaping, giving lower steady-state ripple at comparable transient response. Derived a closed-form stability criterion for gain-scheduled dc–dc converters under pulsed loads that tracks the exact threshold within 13% across a 19× range, demonstrated on a 100 V-to-24 V buck prototype. Developed solid-state Marx generator drivers, a flat-top high-current compensator for magnetic confinement fusion, and power electronic solutions for dielectric barrier discharge (DBD) applications.
- **AI for power electronics:** Developed an EMI-robust physics-informed neural network (PINN) that cuts series-resistance identification error from 96% to 11%, and a spiking neural network (SNN) estimator that cuts it from 25.8% to 10.2% at a projected 270× energy reduction on neuromorphic hardware.
- **Device characterization:** Designed a four-quadrant double-pulse test (DPT) circuit and prototype for dynamic on-resistance characterization of monolithic bidirectional GaN switches (MBDSs).

Design & Control Laboratory, KoreaTech — Cheonan, Korea
Research Assistant (Advisor: *Prof. Ye gu Kang*) — Apr 2022 - Jun 2023

- Developed linear Hall-effect sensor prototypes for x-y and rotary position estimation, designed optimal IPM rotor geometry via differential evolution, and evaluated a dual-mode machine (generator/motor) for maritime applications (Hyosung).

Communication Signal Processing Laboratory, Yonsei University — Seoul, Korea
Research Intern (Advisor: *Prof. Chungyong Lee*) — Jan 2022 – Mar 2022

- Evaluated autoencoder-based deep learning communication system performance and analyzed OFDM modulation and BER over multi- and single-path channels.

TEACHING EXPERIENCE

ECE 712, University of Wisconsin - Madison — Madison, WI
Teaching Assistant (Supervisor: *Prof. Jinia Roy*) — Spring 2026

ECE 412, University of Wisconsin - Madison — Madison, WI
Teaching Assistant (Supervisor: *Prof. Jinia Roy*) — Fall 2025, Fall 2026

- Developed active-learning materials designed for flipped-classroom instruction.
- Led in-class discussions and addressed student questions during lectures and office hours.

TUTORIALS & INVITED PRESENTATIONS

Hyeongmeen Baik and J. Roy, "Active Capacitor Circuit and Control Techniques for Transportation Electrification Systems," 90-min tutorial (peer-reviewed), *2026 IEEE Transportation Electrification Conference & Expo + IEEE Electric Aircraft Technologies Symposium (ITEC+EATS)*, Novi, MI, Jun 2026.

HONORS AND AWARDS

Novotny Dissertator Award, University of Wisconsin - Madison	2026
Best Presentation Award, APEC 2026	Mar 2026
Scholarship for Academic Excellence (Full-tuition), Yonsei University	Fall-2022, Spring-2021
Scholarship for Academic Excellence (Half-tuition), Yonsei University	Spring-2022, Fall-2021
Highest Honors (Top 1% of department), Yonsei University	Spring-2022, Fall-2020
High Honors (Top 3% of department), Yonsei University	Spring-2021, Fall-2019
Honors (Top 10% of department), Yonsei University	Fall-2021

PUBLICATIONS

Hyeongmeen Baik and J. Roy, "Review of Power Electronic Solutions for Dielectric Barrier Discharge Applications," *IEEE Transactions on Power Electronics*, 2026.

Hyeongmeen Baik and J. Roy, "EMI-Robust Physics-Informed Neural Networks for DC-DC Converter Parameter Identification," *2026 IEEE Energy Conversion Congress and Exposition (ECCE)*, 2026 (accepted, to appear).

Hyeongmeen Baik, S. Bae, and J. Roy, "Modeling and Dual-Mode Control of a Pulsed Buck Flashlamp Driver," *2026 IEEE Energy Conversion Congress and Exposition (ECCE)*, 2026 (accepted, to appear).

Hyeongmeen Baik, H. Poursiami, M. Parsa, and J. Roy, "Neuromorphic Parameter Estimation for Power Converter Health Monitoring Using Spiking Neural Networks," *2026 International Conference on Neuromorphic Systems (ICONS)*, 2026 (accepted, to appear). arXiv preprint arXiv:2604.15714, 2026.

S. Kapse, **Hyeongmeen Baik**, and J. Roy, "Design of High-Frequency, High-Voltage Planar Transformer for Driving Dielectric Barrier Discharge Loads," *2026 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Session D31.8, Mar 2026.

H. Kong, **Hyeongmeen Baik**, and J. Roy, "Double-Pulse Test Circuit for Four-Quadrant Characterization and Dynamic ON-Resistance Evaluation of Monolithic Bidirectional GaN Switches," *2026 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Session T22.2, Mar 2026.

Hyeongmeen Baik, I. Prabhu, and J. Roy, "Multi-Port Extended-Duty-Ratio Boost Converter for Hybrid Power Sources," *2025 IEEE Energy Conversion Congress and Exposition (ECCE)*, 2025, pp. 1-6.

P. Javidi, **Hyeongmeen Baik**, and J. Roy, "A Synchronous Gate Driver System for Series-Connected SiC MOSFETs for High-Voltage Pulsed Power Supplies Based on Magnetic Pulse Compression," *2025 IEEE Energy Conversion Congress and Exposition (ECCE)*, 2025, pp. 1-8.

A. Chinnusamy, **Hyeongmeen Baik**, and J. Roy, "High Power Density Solid State Marx Generator Based Pulsed Power Driver for Nonlinear Transmission Line," *2025 IEEE Energy Conversion Congress & Exposition Asia (ECCE-Asia)*, 2025, pp. 1-6.

Hyeongmeen Baik and J. Roy, "Modular Flat-Top High-Current Compensator for Flywheel Energy Storage System in Magnetic Confinement Fusion Applications," *2024 IEEE Energy Conversion Congress and Exposition (ECCE)*, Phoenix, AZ, USA, 2024, pp. 7352-7358, doi: 10.1109/ECCE55643.2024.10861038.

PREPRINTS / UNDER REVIEW

Hyeongmeen Baik, J. M. Eckert, C. Stiemke, L. Mastropasqua, and J. Roy, "Orthonormal Current-Sharing Control of an Interleaved Buck Converter for In-Situ Electrolyzer Characterization via DC-Biased AC Excitation," submitted to *IEEE Transactions on Power Electronics* (under review), 2026.

Hyeongmeen Baik and J. Roy, "Parametric Instability of Gain-Scheduled DC–DC Converters under Pulsed Loads: Monodromy Analysis and Design Guideline," submitted to *IEEE Transactions on Industrial Electronics* (under review), 2026.